

Project 1

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(a)

Monthly adjusted close prices were downloaded from Yahoo Finance through the Shiny app, covering January 2017 through December 2021. Five sectors, six stocks each.

Sector	Tickers
Technology	NVDA, AAPL, MSFT, AVGO, AMD, NOW
Communication Services	GOOG, META, NFLX, VZ, DIS, T
Financial Services	V, JPM, MS, GS, BAC, SCHW
Consumer Cyclical	AMZN, TSLA, MCD, SBUX, HD, EBAY
Healthcare	LLY, NVO, UNH, JNJ, AMGN, PFE

$\hat{GSPC}$ is carried alongside as the market index as asked for, for 31 assets in total.

(b)

Adjusted close prices are converted to simple (arithmetic) monthly returns:

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

```
#> prices : 60 months x 31 assets
```

```
#> returns: 59 months x 31 assets (2017-02-01 to 2021-12-01)
```

The first month is lost to differencing, so we have 59 monthly returns.

Table 1: First four rows (first 7 assets)

$\hat{GSPC}$	NVDA	AAPL	MSFT	AVGO	AMD	NOW
0.0372	-0.0706	0.1336	-0.0043	0.0570	0.3944	-0.0408
-0.0004	0.0720	0.0487	0.0295	0.0429	0.0062	0.0063
0.0091	-0.0410	0.0000	0.0394	0.0082	-0.0859	0.0806

$\hat{G}SPC$	NVDA	AAPL	MSFT	AVGO	AMD	NOW
0.0116	0.3852	0.0678	0.0259	0.0845	-0.1586	0.1074

(c)

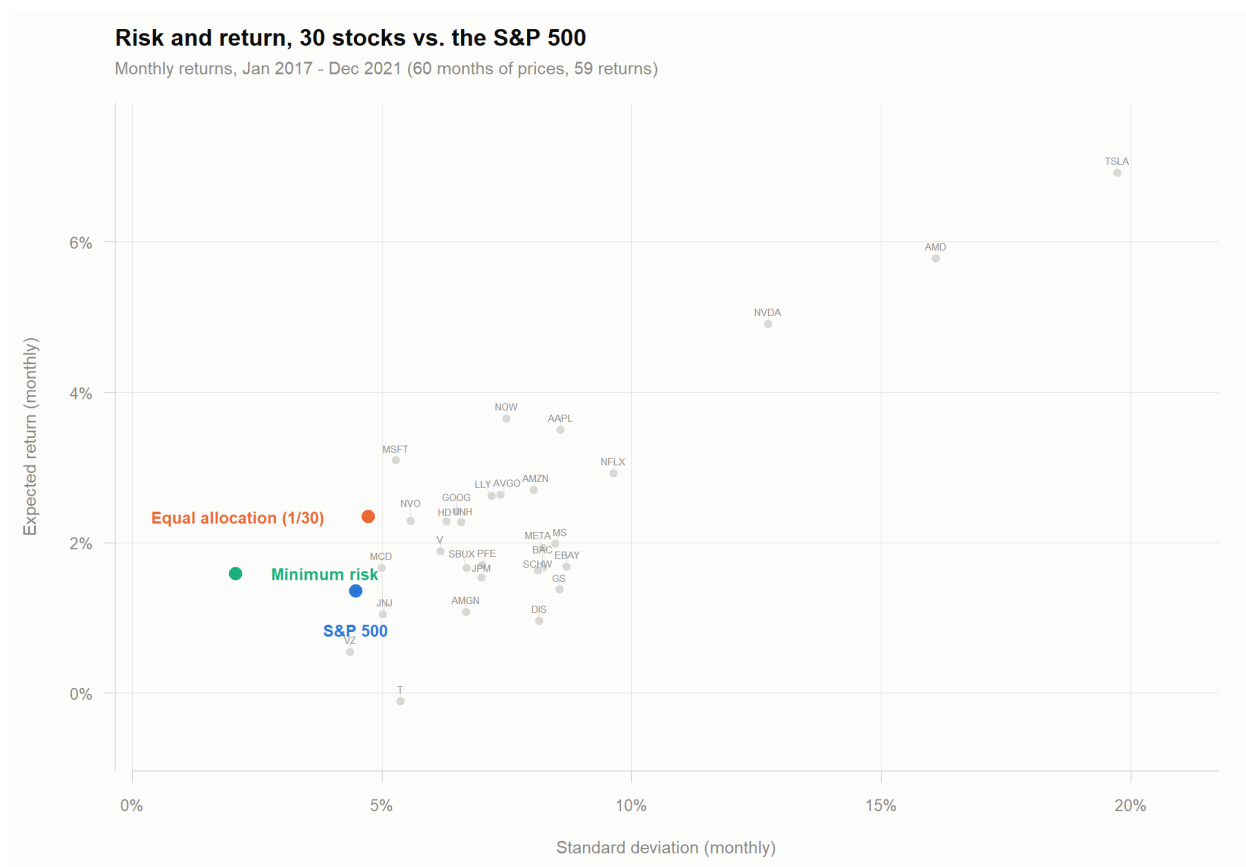
Table 2: All 31 assets, monthly

Asset	Mean (%)	Std. dev. (%)
$\hat{G}SPC$	1.358	4.480
NVDA	4.910	12.740
AAPL	3.503	8.573
MSFT	3.094	5.280
AVGO	2.636	7.372
AMD	5.780	16.102
NOW	3.655	7.497
GOOG	2.414	6.492
META	1.938	8.217
NFLX	2.921	9.633
VZ	0.553	4.357
DIS	0.960	8.143
T	-0.107	5.367
V	1.888	6.172
JPM	1.535	6.988
MS	1.986	8.469
GS	1.381	8.549
BAC	1.666	8.224
SCHW	1.635	8.119
AMZN	2.706	8.044
TSLA	6.919	19.732
MCD	1.667	4.991
SBUX	1.666	6.694
HD	2.282	6.292
EBAY	1.679	8.695
LLY	2.621	7.194
NVO	2.290	5.576
UNH	2.272	6.589
JNJ	1.050	5.012
AMGN	1.077	6.686
PFE	1.704	7.005

The variance-covariance matrix is 31×31 ; the top-left corner is shown below. The full matrix is too wide to display at a readable size, and is computed in the appendix code.

	$\hat{G}SPC$	NVDA	AAPL	MSFT	AVGO	AMD
$\hat{G}SPC$	0.00201	0.00263	0.00241	0.00173	0.00193	0.00376
NVDA	0.00263	0.01620	0.00617	0.00370	0.00319	0.00778
AAPL	0.00241	0.00617	0.00735	0.00275	0.00305	0.00601
MSFT	0.00173	0.00370	0.00275	0.00279	0.00169	0.00334
AVGO	0.00193	0.00319	0.00305	0.00169	0.00543	0.00231
AMD	0.00376	0.00778	0.00601	0.00334	0.00231	0.02590

(d)



Parts (e) and (f) are marked on the same plot.

(e)

Equal weights across the 30 stocks, $w_i = 1/30$. Portfolio mean is $\bar{R}_p = w' \bar{R}$ and standard deviation is $\sigma_p = \sqrt{w' \Sigma w}$.

```
#> mean = 0.0234 (2.34% per month)
#> sd = 0.0473 (4.73%)
```

(f)

The weights of the minimum risk portfolio are

$$x = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}}$$

where Σ is the 30×30 variance-covariance matrix of the 30 stocks and $\mathbf{1}$ is a 30×1 vector of ones.

```
#> mean = 0.0159 (1.59% per month)
#> sd = 0.0207 (2.07%)
#>
#> weights sum to 1.0000, range -0.545 to 0.839, 14 short positions
```

Table 3: Five largest and five smallest weights

Stock	Weight
JPM	0.8395
NFLX	0.2124
VZ	0.1992
NVO	0.1949
AAPL	0.1699
AMD	-0.0914
V	-0.1033
JNJ	-0.2493
MS	-0.2927
BAC	-0.5447

Appendix: full code

```
# Portfolio project, parts (b)-(f)
# Monthly adjusted close prices, first 5 years only (2017-01 .. 2021-12).

local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "portfolio.R")
    setwd(dirname(normalizePath(f)))
})

DATA <- ".."

# ---- (b) import prices and convert to returns -----

# check.names = FALSE keeps "^GSPC" intact instead of mangling it to "X.GSPC".
a <- read.csv(file.path(DATA, "stockData_train.csv"), sep = ",", header = TRUE,
              check.names = FALSE)

stopifnot(!any(duplicated(names(a))))

prices <- a[, -1] # drop the leading row-index column

dates <- as.Date(prices$Date)
P <- as.matrix(prices[, -1]) # 60 x 31, assets in columns

stopifnot(nrow(P) == 60) # 5 years of monthly observations
stopifnot(ncol(P) == 31) # ^GSPC + 30 stocks
stopifnot(!anyNA(P))

# Simple (arithmetic) returns: R_t = P_t / P_{t-1} - 1.
n <- nrow(P)
ret <- P[-1, ] / P[-n, ] - 1 # 59 x 31

cat("(b) prices ", nrow(P), "x", ncol(P),
     " -> returns", nrow(ret), "x", ncol(ret), "\n")
cat("    return period:", format(dates[2]), "to", format(dates[n]), "\n\n")
```

```

# ---- (c) means, standard deviations, variance-covariance matrix ----

means  <- colMeans(ret)           # all 31 assets
covmat <- cov(ret)
stdev  <- sqrt(diag(covmat))

cat("(c) mean monthly return and sd, all 31 assets:\n")
print(round(data.frame(mean_pct = means * 100, sd_pct = stdev * 100), 3))
cat("\n    covariance matrix:", nrow(covmat), "x", ncol(covmat), "\n\n")

# The 30 stocks alone (drop ^GSPC) - this is what parts (e) and (f) use.
stk    <- setdiff(colnames(ret), "^GSPC")
ret30  <- ret[, stk]
mu30   <- colMeans(ret30)
S30    <- cov(ret30)              # 30 x 30
stopifnot(ncol(S30) == 30)
stopifnot(qr(S30)$rank == 30)    # invertible, so part (f) is well posed

# ---- (e) equal allocation portfolio (30 stocks) ----

w_eq   <- rep(1 / 30, 30)
mu_eq  <- as.numeric(t(w_eq) %*% mu30)
sd_eq  <- sqrt(as.numeric(t(w_eq) %*% S30 %*% w_eq))

cat("(e) equal allocation portfolio (w = 1/30 each):\n")
cat(sprintf("    mean = %.4f (%.2f%% / month)    sd = %.4f (%.2f%%)\n\n",
            mu_eq, mu_eq * 100, sd_eq, sd_eq * 100))

# ---- (f) minimum risk portfolio ----
#  $x = (S^{-1} 1) / (1' S^{-1} 1)$ 

one    <- rep(1, 30)
Sinv   <- solve(S30)
x_min  <- Sinv %*% one / as.numeric(t(one) %*% Sinv %*% one)
mu_min <- as.numeric(t(x_min) %*% mu30)
sd_min <- sqrt(as.numeric(t(x_min) %*% S30 %*% x_min))

stopifnot(abs(sum(x_min) - 1) < 1e-10) # weights sum to 1 by construction

cat("(f) minimum risk portfolio:\n")
cat(sprintf("    mean = %.4f (%.2f%% / month)    sd = %.4f (%.2f%%)\n",
            mu_min, mu_min * 100, sd_min, sd_min * 100))
cat(sprintf("    weights sum to %.6f, range %.3f to %.3f, %d short positions\n",
            sum(x_min), min(x_min), max(x_min), sum(x_min < 0)))
cat("    largest 5 weights:\n")
print(round(sort(x_min[, 1], decreasing = TRUE)[1:5], 4))
cat("\n\n")

# ---- (d) plot: expected return vs standard deviation ----

surface <- "#fcbfcb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_gspc <- "#2a78d6"; c_eq <- "#eb6834"; c_min <- "#1baf7a"

```

```

repel <- function(x, y, lab, cex, fixed, iter = 600) {
  w <- strwidth(lab, cex = cex) * 1.08
  h <- strheight(lab, cex = cex) * 1.85
  lx <- x; ly <- y + h * 0.85
  ly[fixed] <- y[fixed]
  for (it in seq_len(iter)) {
    moved <- FALSE
    for (i in seq_along(lx)) for (j in seq_along(lx)) if (i < j) {
      if (fixed[i] && fixed[j]) next
      dx <- lx[j] - lx[i]; dy <- ly[j] - ly[i]
      ox <- (w[i] + w[j]) / 2 - abs(dx)
      oy <- (h[i] + h[j]) / 2 - abs(dy)
      if (ox > 0 && oy > 0) {
        moved <- TRUE
        # split the correction, or give it all to the free one of the pair
        fi <- if (fixed[i]) 0 else if (fixed[j]) 1 else 0.5
        if (oy / h[i] < ox / w[i]) {
          s <- if (dy >= 0) oy else -oy
          ly[i] <- ly[i] - s * fi; ly[j] <- ly[j] + s * (1 - fi)
        } else {
          s <- if (dx >= 0) ox else -ox
          lx[i] <- lx[i] - s * fi; lx[j] <- lx[j] + s * (1 - fi)
        }
      }
    }
    if (!moved) break
  }
  list(x = lx, y = ly)
}

png("risk_return.png", width = 2100, height = 1450, res = 190)
op <- par(bg = surface, mar = c(4.6, 4.8, 4.2, 1.8), family = "sans")

# Pad left and right so the highlight labels have room to sit beside their marks
xr <- c(min(c(stdev, sd_min)) - 0.016, max(stdev) + 0.012)
yr <- range(c(means, mu_eq, mu_min)) + c(-0.006, 0.006)

plot(NA, xlim = xr, ylim = yr, xlab = "", ylab = "", axes = FALSE)
abline(h = pretty(yr), v = pretty(xr), col = grid, lwd = 1)
axis(1, col = "#c3c2b7", col.axis = muted, cex.axis = 0.85, lwd = 1,
     at = pretty(xr), labels = sprintf("%.0f%%", pretty(xr) * 100))
axis(2, col = "#c3c2b7", col.axis = muted, cex.axis = 0.85, lwd = 1, las = 1,
     at = pretty(yr), labels = sprintf("%.0f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.9, col = muted, cex = 0.92)
mtext("Expected return (monthly)", 2, line = 3.4, col = muted, cex = 0.92)
mtext("Risk and return, 30 stocks vs. the S&P 500", 3, line = 2.4,
     adj = 0, col = ink, cex = 1.22, font = 2)
mtext("Monthly returns, Jan 2017 - Dec 2021 (60 months of prices, 59 returns)",
     3, line = 1.1, adj = 0, col = muted, cex = 0.9)

# The 30 individual stocks recede - they are context, not the finding.
is_g <- colnames(ret) == "^GSPC"
px <- c(stdev[!is_g], stdev[is_g], sd_eq, sd_min)

```

```

py <- c(means[!is_g], means[is_g], mu_eq, mu_min)
pl <- c(colnames(ret)[!is_g], "S&P 500", "Equal allocation (1/30)",
       "Minimum risk")
pf <- c(rep(FALSE, 30), TRUE, TRUE, TRUE)
pc <- c(rep(0.52, 30), 0.86, 0.86, 0.86)

points(px[1:30], py[1:30], pch = 21, bg = "#d9d8d2",
       col = surface, lwd = 1.4, cex = 1.25)

# Pin the highlight labels beside their marks, then let tickers repel around them
lp <- list(x = px, y = py)
lp$x[31] <- px[31]; lp$y[31] <- py[31] - 0.0052 # S&P 500, below
lp$x[32] <- px[32] - 0.0088; lp$y[32] <- py[32] # equal alloc, left
lp$x[33] <- px[33] + 0.0072; lp$y[33] <- py[33] # min risk, right
lp <- repel(px, py, pl, pc, pf)
lp$x[31:33] <- c(px[31], px[32] - 0.0088, px[33] + 0.0072)
lp$y[31:33] <- c(py[31] - 0.0052, py[32], py[33])

# Leader lines where a ticker drifted away from its point
d <- sqrt(((lp$x[1:30] - px[1:30]) / diff(xr))^2 +
          ((lp$y[1:30] - py[1:30]) / diff(yr))^2)
seg <- which(d > 0.018)
segments(px[seg], py[seg], lp$x[seg], lp$y[seg], col = "#dcd8d4", lwd = 0.8)
text(lp$x[1:30], lp$y[1:30], pl[1:30], cex = 0.52, col = muted)

# Three highlighted entities, each with a surface ring and a direct label.
for (k in 31:33) {
  col <- c(c_gspc, c_eq, c_min)[k - 30]
  points(px[k], py[k], pch = 21, bg = col, col = surface, lwd = 2.4, cex = 2.0)
  text(lp$x[k], lp$y[k], pl[k], cex = 0.86, font = 2, col = col,
       adj = c(if (k == 32) 1 else if (k == 33) 0 else 0.5, 0.5))
}

par(op); dev.off()
cat("(d) wrote risk_return.png\n")

# ---- save tables -----

write.csv(data.frame(Date = dates[-1], ret, check.names = FALSE),
          "returns_train.csv", row.names = FALSE)
write.csv(data.frame(asset = names(means), mean = means, sd = stdev),
          "asset_stats.csv", row.names = FALSE)
write.csv(covmat, "covmat.csv")
cat("    wrote returns_train.csv, asset_stats.csv, covmat.csv\n")

```